Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **Python backends**, **TypeScript/JavaScript interfaces**, and **AI-powered applications** across AWS-backed systems, with experience in rapid prototyping, data pipelines, and customer-facing delivery in fast-paced environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning SQL access patterns, and improving responsiveness for analytics workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures and accelerating incident triage through earlier detection of regressions, data-quality drift, and runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable backend services and responsive interfaces while improving delivery predictability in a fast-moving startup.
- Built **AI-assisted workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production-scale operations and iterating rapidly based on real-world usage patterns.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **large-scale experimentation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation and high-confidence benchmarking while improving signal quality for model-driven decision workflows.
- Produced **technical reports and test scenarios** translating findings into actionable guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward deployment.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, resilient data movement, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for high-volume processes.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Skills

Languages: **Python**, **TypeScript**, **JavaScript**, SQL, Java, C++, Bash
Frontend: React, Next.js, component-driven interfaces, responsive web workflows
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, ETL pipelines, distributed workflows
Cloud & Infrastructure: AWS (EC2, S3, Glue, ECS), Docker, Kubernetes, Linux, CI/CD, Git
AI/ML: PyTorch, Transformers, LLM integration, data pipelines, model evaluation workflows

Projects

PiXMod Live | *Next.js, MongoDB, Transformers, CLIP, SAM*
- Built an **AI-powered Next.js platform** for image processing and search, integrating ML models (CLIP, SAM) with MongoDB-backed workflows and improving premium conversion by **18%** through polished user experiences.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring and schema exploration through a unified Java and TypeScript codebase with Docker-backed development workflows.